Final Project Proposal

Interpretable Sentiment Analysis of Customer Reviews Using Deep Learning

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Problem Definition

Millions of customer reviews are posted online every day across platforms such as Amazon and Yelp. For businesses, these reviews represent an invaluable source of feedback about their products and services. However, manually reading and analysing thousands of reviews is impossible at scale. A small business owner with 500 reviews, a product manager tracking a new launch, or a customer service team monitoring complaints - all face the same challenge: understanding what customers truly think, quickly and accurately. Traditional NLP approaches such as rule-based classifiers and bag-of-words models can label reviews as positive or negative, but they fail to capture context, sarcasm, and nuanced language. More critically, they cannot explain WHY a review was classified a certain way. A business does not just need to know that a review is negative - it needs to know whether customers are complaining about delivery, product quality, or customer service. The primary contribution of this project is a comparative study of four architectures - from traditional ML to state-of-the-art transformers - with an interpretability layer via attention visualization and cross-domain robustness analysis on Yelp reviews.

Aim and Objectives

The aim of this project is to develop and evaluate a progressive deep learning pipeline for sentiment classification that is both accurate and interpretable.

Objectives:

1. Compare four model architectures: Logistic Regression, Feedforward Neural Network, LSTM, and DistilBERT - on the Amazon customer reviews dataset.

2. Fine-tune DistilBERT (distilbert-base-uncased) on a stratified subset [6] of Amazon reviews and evaluate its performance against simpler baselines.

3. Generate attention visualizations for DistilBERT and extract LR coefficients to provide an interpretability comparison across model complexity levels.

4. Evaluate cross-domain generalization by testing the best performing model on Yelp reviews- a different platform but identical binary sentiment task.

Data Sources

This project uses two datasets - the Amazon Reviews dataset (split into 200,000 training and 40,000 test samples) and the Yelp Reviews dataset (5,000 samples used exclusively for cross-domain generalization testing).

Amazon Reviews (train.ft.txt)	Kaggle [1]	3.6M total Sample: 200000	Binary labels: __label__1 (negative), __label__2 (positive). Balanced 50/50. Mean length: 78 words, median: 70 words.
Amazon Reviews (test.ft.txt)	Kaggle [1]	400000 total Sample: 40000	Same format as training set. Used for standard model evaluation.
Yelp Reviews (cross-domain)	Kaggle [2]	38 total Sample: 5000	Binary labels: 0 (negative), 1 (positive). Balanced 50/50. Mean length: 133 words, median: 97 words.

All datasets are publicly available, contain no personally identifiable information, and raise no ethical concerns. Due to computational constraints, stratified samples are used- maintaining class balance across all subsets. Zero missing values were found across all three datasets.

Proposed Approach

The project implements a four-stage comparative pipeline. Preprocessing differs across models - heavy preprocessing (lowercasing, stop-word removal, punctuation removal) is applied for Logistic Regression and FNN, minimal preprocessing for LSTM, and no stop-word removal or lowercasing for DistilBERT which uses its own subword tokenizer:

1. <u>Logistic Regression (Baseline):</u> TF-IDF vectorization converts pre-processed text into numerical features. Logistic Regression is trained as a baseline benchmark. Top positive and negative coefficient words are extracted as a simple interpretability analysis.

2. <u>Feedforward Neural Network (FNN):</u> TF-IDF features serve as input to a two-layer feedforward network. TF-IDF features serve as input to a two-layer feedforward network (Dense 256 - Dropout 0.3 - Dense 128 - Dropout 0.3 - Output), with dropout regularization to prevent overfitting.

3. <u>LSTM (Long Short-Term Memory)</u> [5]: Text is tokenized and padded to max length 128. GloVe 100-dimensional pre-trained embeddings initialize the embedding layer. A bidirectional LSTM (hidden size 64) captures sequential dependencies in both directions. Dropout 0.3 applied for regularization.

4. <u>DistilBERT (Transformer):</u> distilbert-base-uncased [4] is loaded via HuggingFace Transformers and fully fine-tuned on a stratified subset of 15,000 Amazon reviews (GPU required). Attention weight visualization generates heatmaps showing which tokens most influenced each prediction [3]. Results are compared against LR coefficients for an interpretability comparison axis.

All models are evaluated on the Amazon test set (40,000 reviews) using Accuracy, Precision, Recall, F1-Score (primary metric), and ROC-AUC. The best performing model is then tested on 5,000 Yelp reviews to evaluate cross-domain generalization. All experiments run in Python using PyTorch and HuggingFace.

Timeline

1	May 11-17	Data collection and exploration	Literature review, load and explore all datasets, confirm class balance and text length statistics.
2	May 18-24	Data preprocessing	Text preprocessing pipelines (separate for LR/FNN vs LSTM vs DistilBERT), TF-IDF feature extraction, label conversion.
3	May 25-31	Baseline models implementation	Implement Logistic Regression and FNN baselines, extract LR coefficients for interpretability. Deliverable: LR + FNN notebooks with baseline F1 scores.
4	Jun 1-7	LSTM development	Implement bidirectional LSTM with GloVe 100d embeddings, tune hyperparameters, compare against baselines. Deliverable: LSTM notebook with evaluation metrics. (Biweekly project check-in).
5	Jun 8-11	DistilBERT Fine-tuning	Fine-tune DistilBERT, generate attention heatmaps, compare all four models. Deliverable: DistilBERT notebook with attention visualizations. (Project progress report and biweekly project check-in).
6	Jun 12-14	Cross-domain testing and model comparison	Cross-domain testing on Yelp reviews, ablation study, finalize model comparison table. Deliverable: Complete results with cross-domain analysis.
7	Jun 15-22	Report and presentation	Final report and presentation. Deliverable: Final report and presentation.

This final proposal was updated following a project check-in on May 21. Professor feedback was incorporated, including the addition of explicit objectives, detailed model architecture specifications, and differentiated preprocessing strategies.

References

[1] Amazon Customer Reviews Dataset. (2023). Kaggle. https://www.kaggle.com/datasets/bittlingmayer/amazonreviews

[2] Yelp Reviews Sentiment Dataset. (2022). Kaggle. https://www.kaggle.com/datasets/thedevastator/yelp-reviews-sentiment-dataset

[3] Vaswani, A., et al. (2017). Attention Is All You Need. Advances in Neural Information Processing Systems, 30. https://arxiv.org/abs/1706.03762

[4] Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. Proceedings of NAACL-HLT 2019. https://arxiv.org/abs/1810.04805

[5] Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735-1780. https://www.bioinf.jku.at/publications/older/2604.pdf

[6] Sanh, V., Debut, L. Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. https://arxiv.org/abs/1910.01108